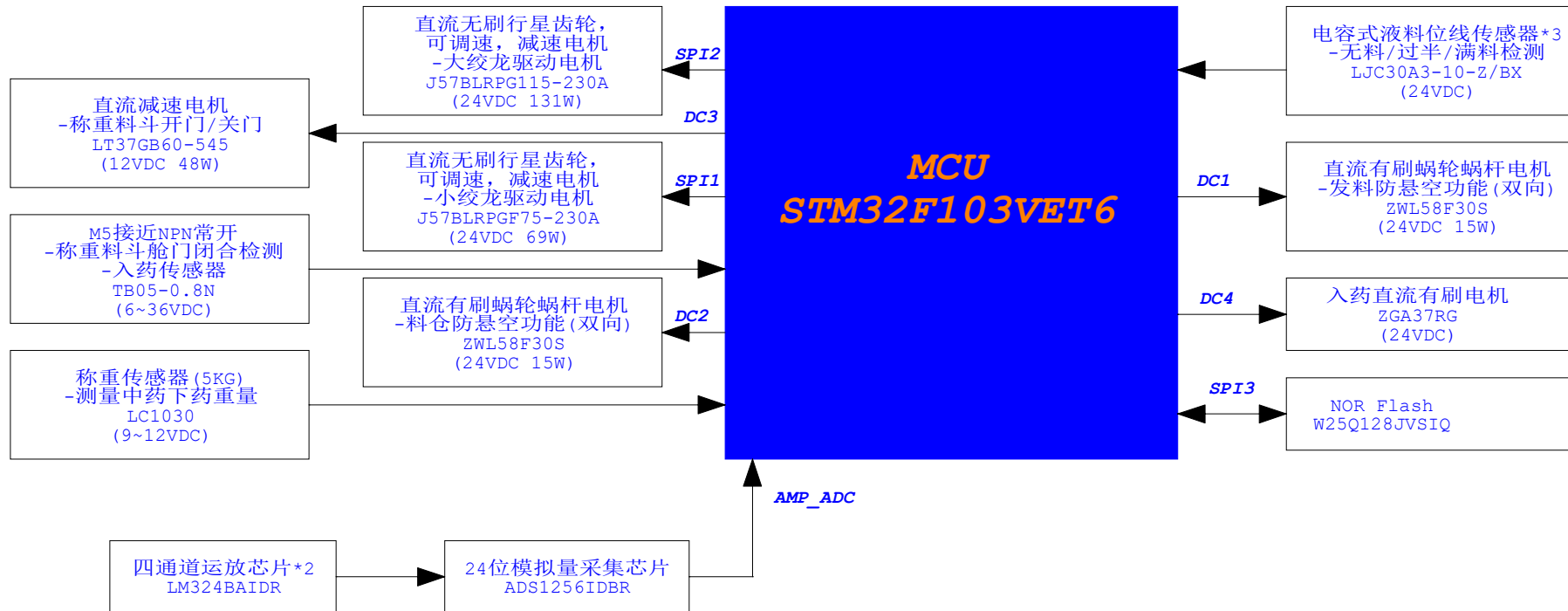
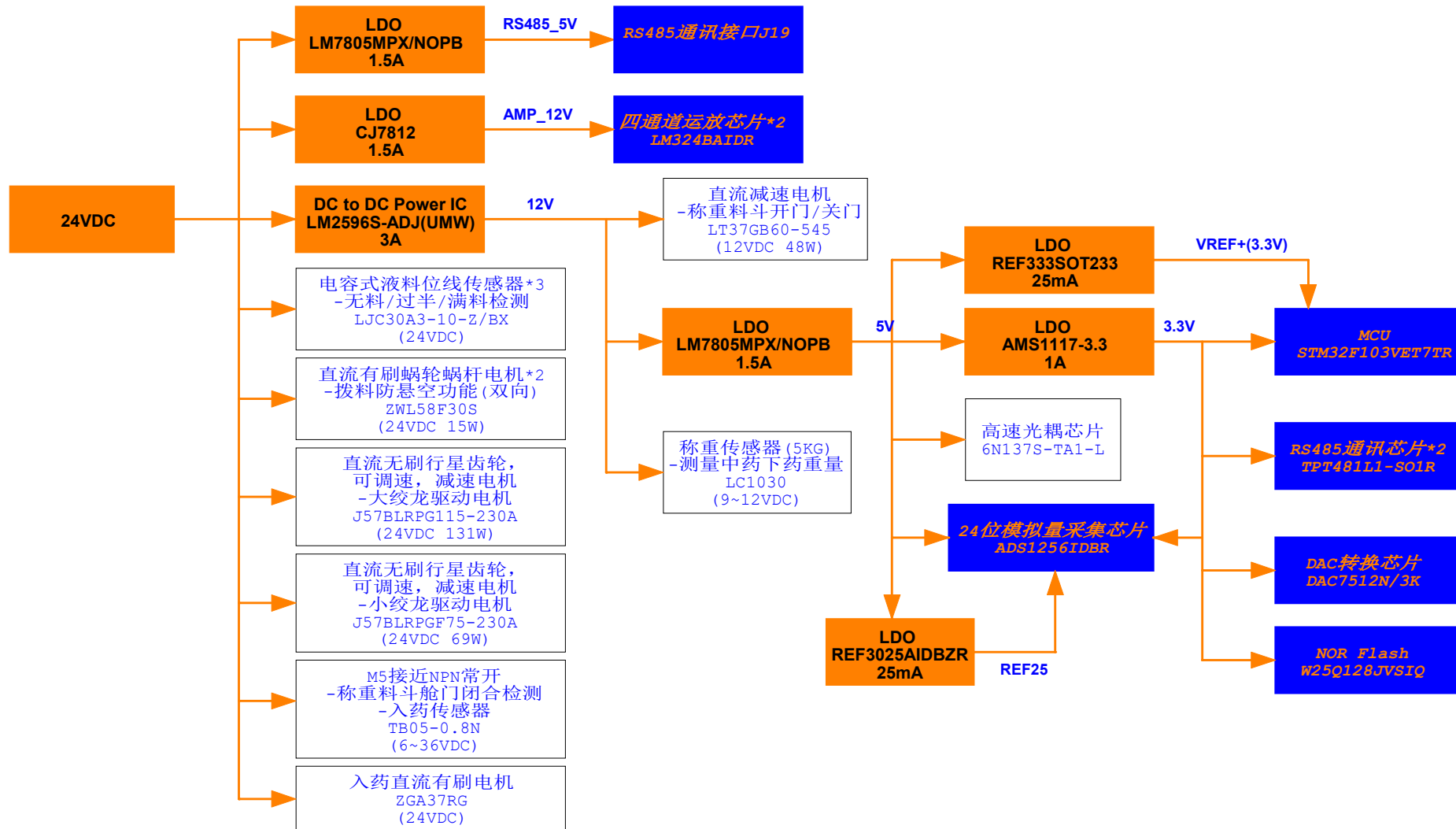


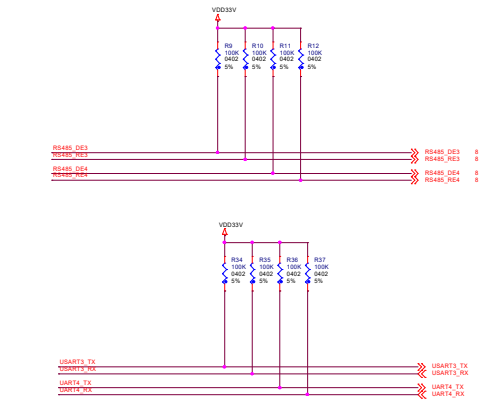
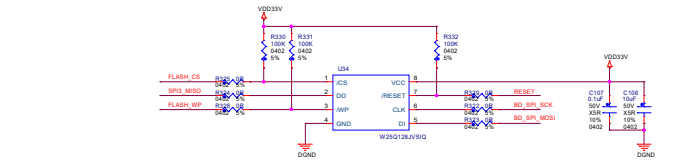
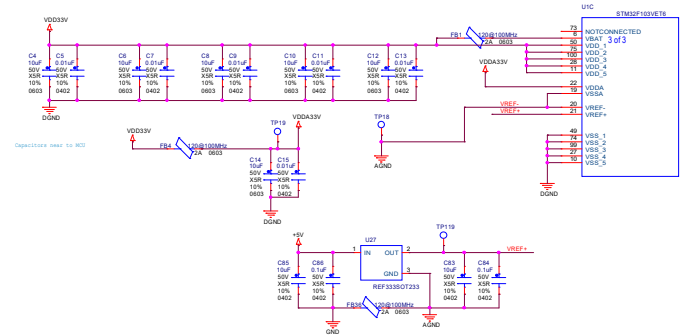
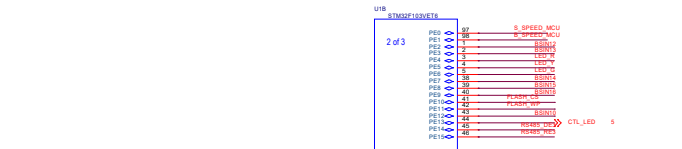
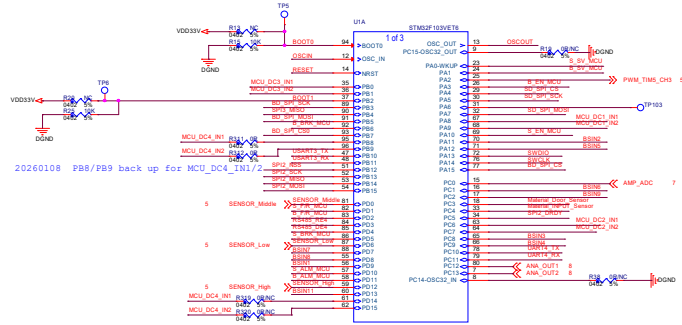
# Function



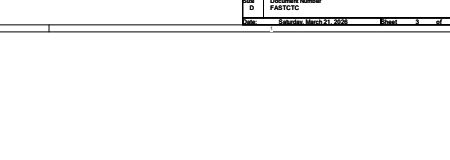
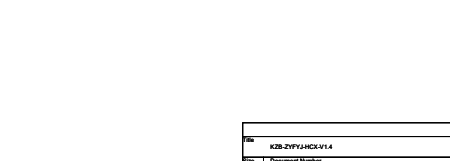
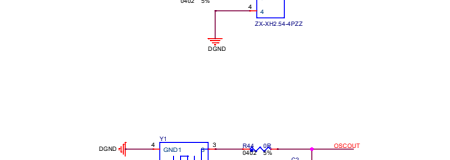
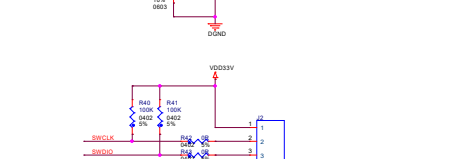
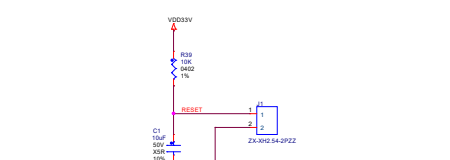
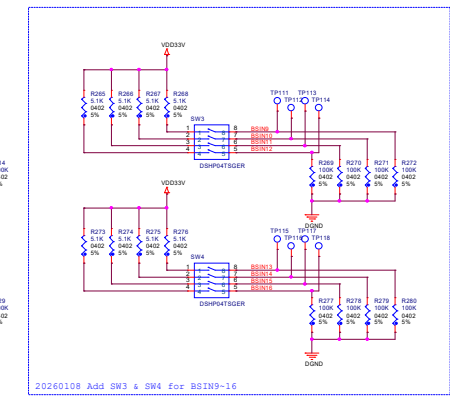
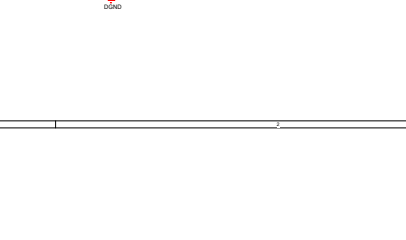
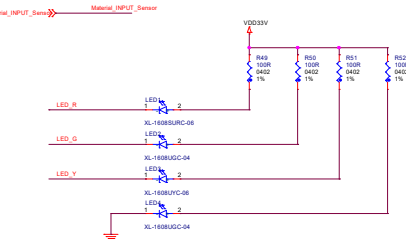
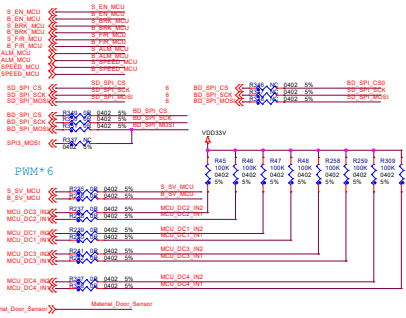
## Power Diagram



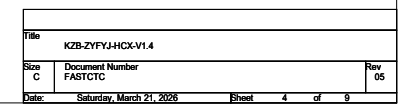
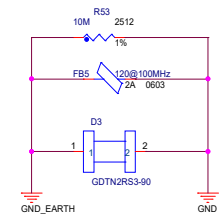
# MCU



| HW design:  | PIN design: | PWM design: |
|-------------|-------------|-------------|
| MCU_DC1_IN1 | PIN67       | PA8         |
| MCU_DC1_IN2 | PIN68       | PA9         |
| MCU_DC2_IN1 | PIN63       | PC6         |
| MCU_DC2_IN2 | PIN64       | PC7         |
| MCU_DC3_IN1 | PIN35       | PB0         |
| MCU_DC3_IN2 | PIN36       | PB1         |
| S_V_MCU     | PIN23       | PA0         |
| B_V_MCU     | PIN24       | PA1         |
| MCU_DC4_IN1 | PIN95       | PB8         |
| MCU_DC4_IN2 | PIN96       | PB9         |

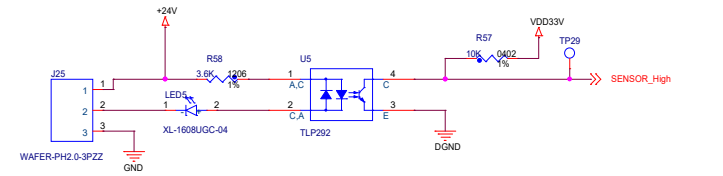


## Power

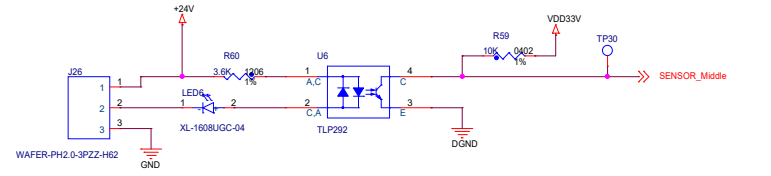


# 料仓传感器+防悬空

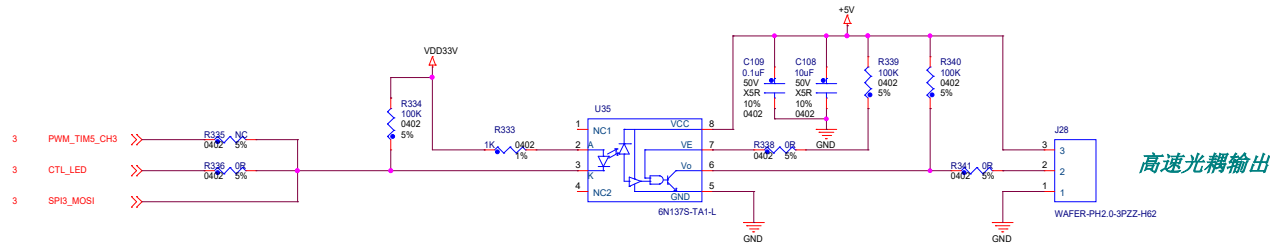
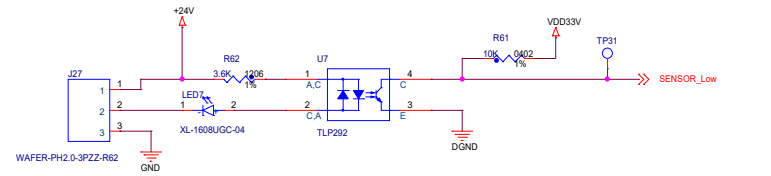
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LIC30A3-10-Z/BX  
(常开-High level)



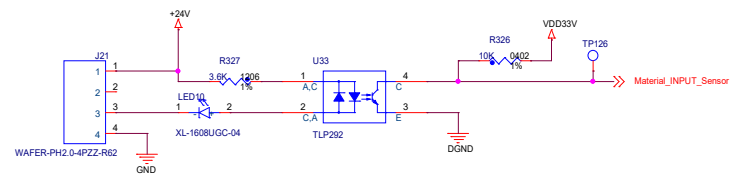
料仓传感器-中  
LIC30A3-10-Z/BX  
(常开-High level)



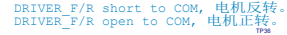
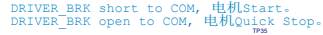
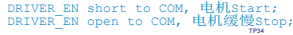
料仓传感器-下  
LIC30A3-10-Z/BX  
(常开-High level)



入药传感器  
TB05-0.8N  
(NPN 常开)

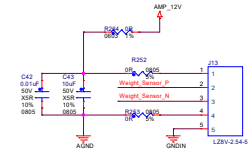
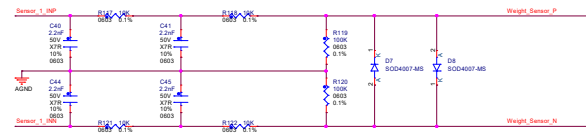


## 发料+防悬空

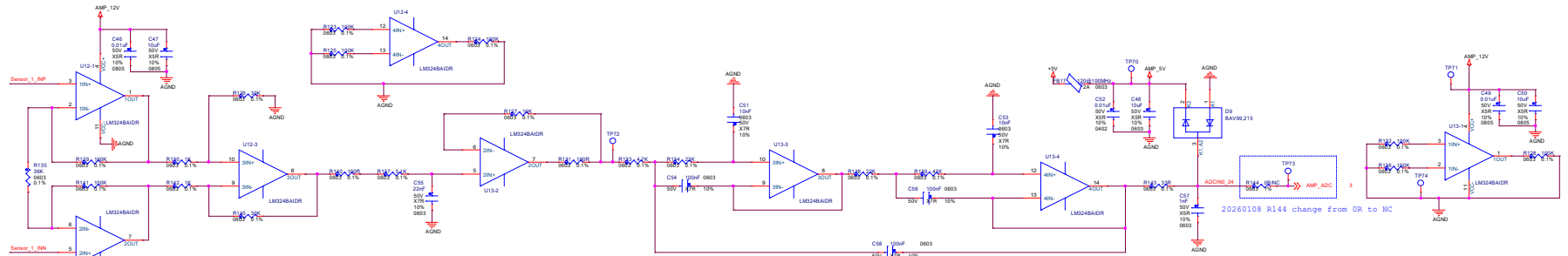
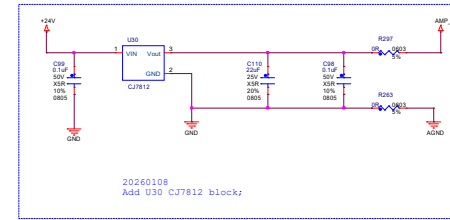


|                    |                          |       |        |
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| File               |                          |       |        |
| KZB-ZYPJA-HCX-V1.4 |                          |       |        |
| Size               | Document Number          |       | Rev    |
| D                  | FASTCTC                  |       | 05     |
| Date:              | Saturday, March 21, 2008 | Sheet | 6 of 2 |

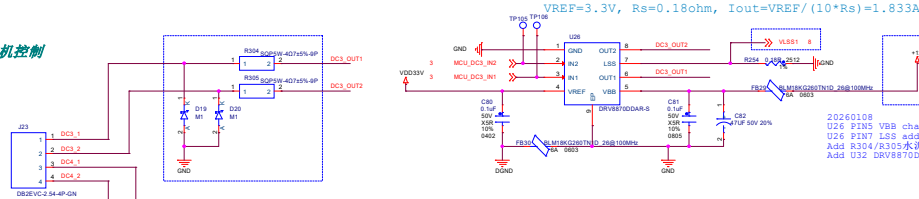
## 称重+料斗门控制&检测



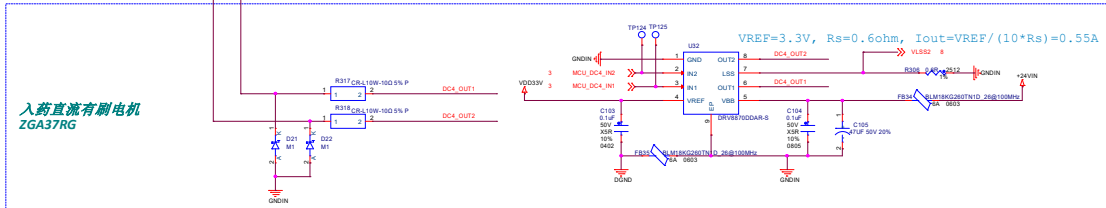
## 称重传感器 LC1030



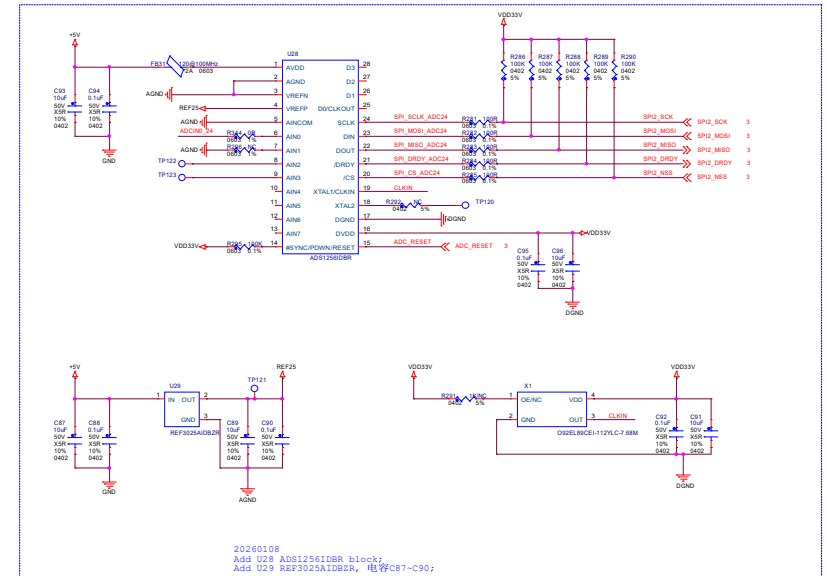
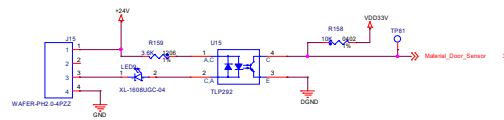
料斗门直流电机控制  
LT37GB60-545



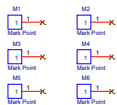
入药直流有刷电机  
ZGA37RG



料仓门关闭检测  
TB05-0.8N  
(NPN常开)

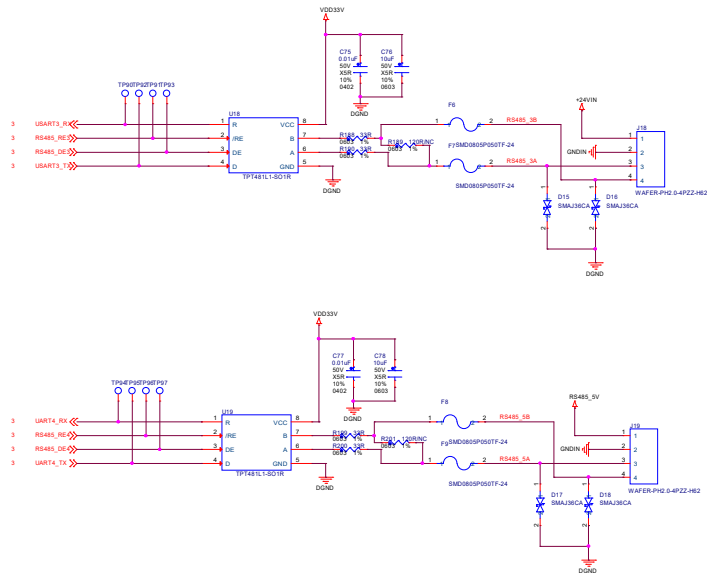
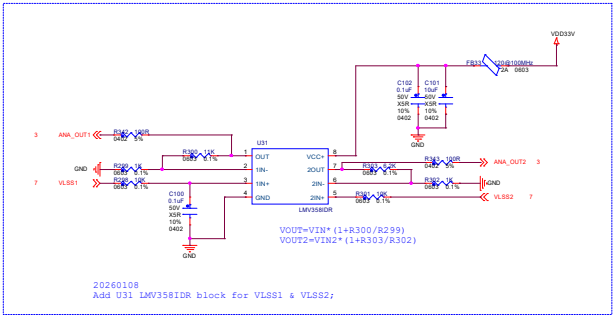


LED & RS485



20260108 删除J20和所有相关输出电路LED1-LED10。

20260108 DELETE U16/U17以及相关电路。



HISTORY

| REV | SCHEMATICS CHANGES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | REV | LAYOUT CHANGES                                                                                                                                                                                                                                                                                                                                            | RELEASE DATE |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| 01  | INITIAL RELEASE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 01  | INITIAL RELEASE                                                                                                                                                                                                                                                                                                                                           | 2025/08/20   |
| 02  | 20250916 update:<br>1. Page1:<br>(1) 直流减速电机更新为12V 48W<br>(2) 电容式液料位线传感器变更为LJC30A3-10-Z/BX (24VDC)<br>2. Page2, History记录更新<br>3. Page3:<br>(1) 连接到page6的S_DRIVER_CTL和B_DRIVER_CTL控制信号取消, 改为测试点。<br>(2) 连接到page7的MCU_DC3_CLK和DC3_PWM_CTL控制信号取消。<br>(3) S_SV_MCU 改为连接MCU PIN23_PA0 (TIM5_CH1), B_SV_MCU改为连接MCU PIN24_PA1 (TIM2_CH2)。<br>(4) 新增MCU_DC3_IN1连接至MCU PIN35_PB0 (TIM3_CH3)<br>MCU_DC3_IN2连接至MCU PIN36_PB1 (TIM3_CH4), 增加上拉电阻R258&R259_100K_0402。<br>(5) 连接到page7的DC3_CTL和DC3_SV的ADC信号 (R246、R247) 取消, 改为测试点。<br>(6) SD_SPL_MOS1从PB5改回PA7<br>(7) SENSOR_Middle从PB1改至PB5<br>4. Page5, C17 从0402变更成0603。<br>5. Page6:<br>(1) FB10换成FB26_6A_0603。<br>(2) U8 VBB电压输入变更为+24VIN和FB25_6A_0603。GND和EP连接变更为GNDIN;<br>6. Page6:<br>(1) DRIVER_24V 从+24V&GND换成+24VIN&GNDIN。线宽和线距至少50mil以上。<br>(2) FB20和FB21互换元件型号, 电流差异。<br>(3) S_DRIVER_CTL和B_DRIVER_CTL控制电源取消, 删除Q12/R100/R101和Q14/R114/R115。<br>(4) Driver信号前GND换成DRGND。<br>(5) FB13换成FB28_6A_0603。<br>(6) U9/U10 out 与S_DRIVER_SV之间增加一个AMP电压跟随芯片U25(LM358ADR, SOIC8)。<br>(7) R95/R109 改成NC。<br>(8) R93/R106/R260/R261从620R改为51R。<br>(9) 增加R262连接接DGND与DRGND。<br>(10) U11 VBB电压输入变更为+24VIN和FB27_6A_0603。GND和EP连接变更为GNDIN;<br>7. Page7:<br>(1) FB15&FB16换成R252/R253_0ohm电阻_0805。<br>(2) 电子门直流电机控制, 去除K1_、Q15_、U14_、U24及相关电路; 换成12V输入+H桥芯片DRV8870DDAR (U26)。<br>(3) U26 VBB电压输入变更为+24VIN和FB29_6A_0603。GND和EP连接变更为GNDIN;<br>(4) GNDIN与DGND之间改为FB30_6A_0603。<br>(5) LED10和R160~R163/R165/R168~R172的电路去除, 只保留J20_ J20 pin从LED_CTL换成LED信号。<br>20250918 update<br>1. Page5/6/7:<br>(1) U8/U11/U26 pin7 LSS的接地电阻连接的地从DGND换成GNDIN<br>(2) U8/U11/U26 pin5 VBB的0.1uF电容C29/C38/C81从0402变更为0805。<br>2. Page4:<br>(1) C17从0603换成0805。C19/C23从0402换成0805<br>3. Page7:<br>(1) C42/C46/C49从0402换成0805。C43/C47/C50从0603换成0805 | 02  | 20250916 update:<br>1. 24V、12V与地和其他信号之间隔离距离改到至少30mil以上。                                                                                                                                                                                                                                                                                                   | 2025/09/18   |
| 03  | 20251024<br>1. Page1&2 U1 change from STM32F103VET7 to STM32F103VET6.<br>2. Page4 U2 modify error: Pin3&6 connect to GND.<br>3. Page6 R223&R227 change from 2K to NC.<br>3. Page7 R135 change from 100K to 30K.<br>Add R263(0ohm_0805) between AGND and GND.<br>FB14 change from Bead to R264_OR_0603_1%.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 03  | 20251024 update:<br>1. FB27 & FB25连24V处需要多打过孔。<br>2. FB26 & FB28 & FB30两边的DGND和GNDIN需要增加对地接触面积。                                                                                                                                                                                                                                                           | 2025/10/24   |
| 04  | 20260112<br>1. Page03:<br>(1) Add SW3 & SW4 for BSIN9~16;<br>(2) Delete LED_CTL1~10信号;<br>(3) Delete UART5_RX/UART5_TX/USART2_RX/USART2_TX信号 & 上拉电阻R30~R33;<br>(4) Delete RS485_RE1/RS485_DE1/RS485_RE2/RS485_DE2信号 & 上拉电阻R5~R8;<br>(5) U1_Pin21 VREF+取消VDD33V输入, 删除FB3<br>新增U27_REF33SOT23芯片, 芯片输出连接VREF+, 新增电容(C83~C86);<br>(6) Add MCU_DC4_IN1/2 for PIN61/61, 新增电阻R307~R310;<br>(7) Add Nor Flash U34_W25Q128VSIQ block;<br>(8) Add ANA_OUT1/2 from page8 to PC12/PC13;<br>2. Page05&06<br>(1) 防烧空流直流有刷电机2从page05切换到page06;<br>(2) DC1 & DC2的输出pin增加一根管M1(D22~D26);<br>(3) J7 & J12合并成4PIN螺钉式接线端子J22 DB125-2.54-4P-GN-S;<br>3. Page07<br>(1) R144 change from 0R to NC;<br>(2) Add U28 ADS1256IDBR block;<br>(3) Add U29 REF3025AIDBZR, 电容C87~C90;<br>(4) Add U30 CI7812 block;<br>(5) U26 PIN5 VBB change from +24VIN to +12V;<br>U26 PIN7 LSS add VLSS1 signal connected to Page8;<br>Add R304/R305电源电阻SQP5W-407*5%-9P & D19/D20 二根管M1;<br>(6) Add U32 DRV8870DDAR-S block for ZG&37RG电机驱动(DC4);<br>Add R317/R318电源电阻CR-L10W-10Q 5% P & D21/D22 二根管M1;<br>(7) J14 & DC4新增电机接口合并成4PIN螺钉式接线端子J23 DB125-2.54-4P-GN-S;<br>(8) ADC部分C41/C45 change from NC to 2.2nF, C58 change from NC to 100nF;<br>(9) J13 change from ZX-XH2.54-4PZZ to DB125-2.54-4P-GN-S;<br>4. Page08:<br>(1) Delete J20 & LED1~LED10相关电路(Q16~Q25, R173~R176, R180~R187, R191~R198);<br>(2) Delete U16/U17以及相关电路(J16/J17, F2~F5, D11~D14, C71~C74, R164/R166/R167/R177~R179);<br>(3) Add U31 LMV358IDR block for VLSS1 & VLSS2;                                                                                                                                                                                                                                                     | 04  | 20260112 update:<br>1. 所有电机电路和24V,12V以及相关网络走线, 和其他走线和地铜箔保持净空距离20mil以上。<br>2. DRV8870DDAR-S芯片(U26/U32/U11/U8)芯片layout上为增加散热孔。<br>3. 所有本底电阻(R304/R305/R317/R318)前后的走线与其余走线和地, 请保持20mil以上净空距离。<br>4. 制板时内外层铜厚为2盎司。                                                                                                                                           | 2026/01/12   |
| 05  | 20260321<br>1. Page04: 增加一路24V转5V的电源输出(U36), 和现有电源隔离, 输出RS485_5V接连接器J19的Pin1。<br>J3修改from W1500V-5.08-3P to DB2EVC-5.08-3P-GN。<br>2. Page07: 连接J23修改from DB125-2.54-4P-GN-S to L28V-2.54-5P。<br>连接器J23修改from DB125-2.54-4P-GN-S to DB2EVC-2.54-4P-GN。<br>3. Page06: 连接器J22修改from DB125-2.54-4P-GN-S to DB2EVC-2.54-4P-GN。<br>4. 板名修改成KZB-ZYFYJ-HCX-V1.4。                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 05  | 20260321<br>1. Page04: 增加一路24V转5V的电源输出(U36), 和现有电源隔离, 输出RS485_5V接连接器J19的Pin1。<br>J3修改from W1500V-5.08-3P to DB2EVC-5.08-3P-GN。<br>2. Page07: 连接器J23修改from DB125-2.54-4P-GN-S to L28V-2.54-5P。<br>连接器J23修改from DB125-2.54-4P-GN-S to DB2EVC-2.54-4P-GN。<br>3. Page06: 连接器J22修改from DB125-2.54-4P-GN-S to DB2EVC-2.54-4P-GN。<br>4. 板名修改成KZB-ZYFYJ-HCX-V1.4。 | 2026/03/21   |

|                    |                          |       |        |
|--------------------|--------------------------|-------|--------|
| Title              |                          |       |        |
| KZB-ZYFYJ-HCX-V1.4 |                          |       |        |
| Size               | Document Number          |       | Rev    |
| C                  | FASTCTC                  |       | 05     |
| Date:              | Saturday, March 21, 2026 | Sheet | 9 of 9 |